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Department of Electrical Engineering

End Term Examination December 2022

B.Tech, III SEM, ELECTRICAL ENGINEERING

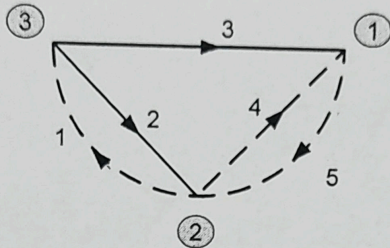
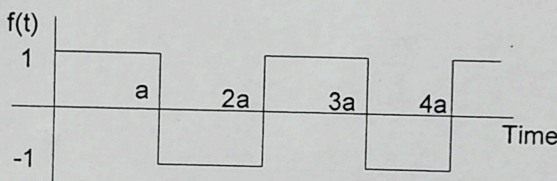
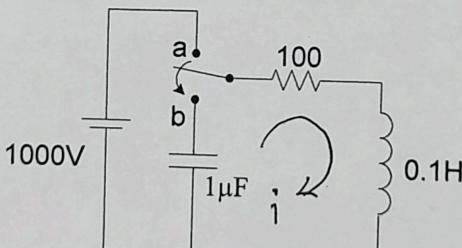
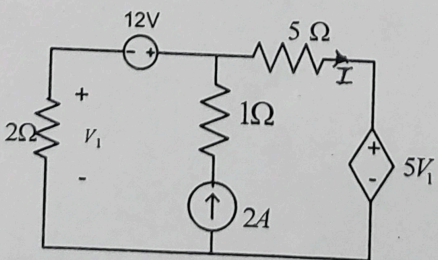
Sub: Network Analysis

Time: 3 hrs.

Sub. Code: EE-213

Max. Marks: 50

Note: Attempt all questions. All questions carry equal marks. Assume suitable data wherever necessary.

Q.1(a)	Write the cutset matrix, KCL and KVL equations for the network shown. Tree branches are marked as bold lines. 	5
(b)	Prove that the Laplace transform of the function shown is $F(s) = \frac{1}{s} \left(\frac{1 - e^{-as}}{1 + e^{-as}} \right)$ 	5
Q.2(a)	For the circuit shown in fig. the switch is moved from a to b at $t = 0$. Find i, $di(0^+)/dt$ and $d^2i(0^+)/dt^2$ at $t = 0^+$ 	5
(b)	Use Superposition theorem to find current I in the fig shown 	5

Q.3(a)	The network shown is initially under steady state condition. The switch is opened at $t=0$. Find the voltage across inductance as a function of t using Laplace transform.	5
(b)	Find current I_2 and voltage V_2 for which current I_1 is zero. Given $X_1=X_2=15\Omega$, and $X_m=10\Omega$.	5
Q.4(a)	Find the Y and z parameters for the network shown.	5
(b)	If the two, two port networks are connected in parallel find the Y parameters of the equivalent network in terms of the Y parameters of the individual networks.	5
Q.5(a)	Synthesize the given function using first Foster form (RC network) $Z(s) = \frac{2(s+2)(s+4)}{(s+1)(s+3)}$	5
(b)	What is the importance of Fourier analysis in Electrical engineering? Explain Half wave symmetry.	5

Sub Coordinator : Dr. Savita Nema